

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems

/ 5G / 6G Communication Systems COURSE

CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Number : CO3_AT2

Assessment Tool Weightage : 35%

Assessment Tool : Design Task

Duration : 180 Minutes Marks

: 100 Marks

Type of Assessment Conduction : At Home

Evaluation Type : GitHub

Result Generation : Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.

Topic Covered :

Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills. This question paper consists of a total of 40 questions, out of which each student is required to answer any 2 questions. Each question carries 50 marks, and the paper carries a total of 100 marks

QUESTIONS: 2 (2 X 50 = 100) Total Marks: 100 Duration: 3hrs

4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)

Code of Conduct:

I K. Dinesh Kumar/192312665L (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and welljustified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided

Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation
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4. Propose a cloud-based wireless framework for smart surveillance systems

Aim:

To design and simulate a cloud-based wireless smart surveillance framework using 5G/6G communication, where surveillance cameras transmit video/data wirelessly to a cloud server with high data rate, low latency, and reliable connectivity. The system performance is evaluated under different SNR conditions using throughput, latency, and packet-error performance.

Program Code:

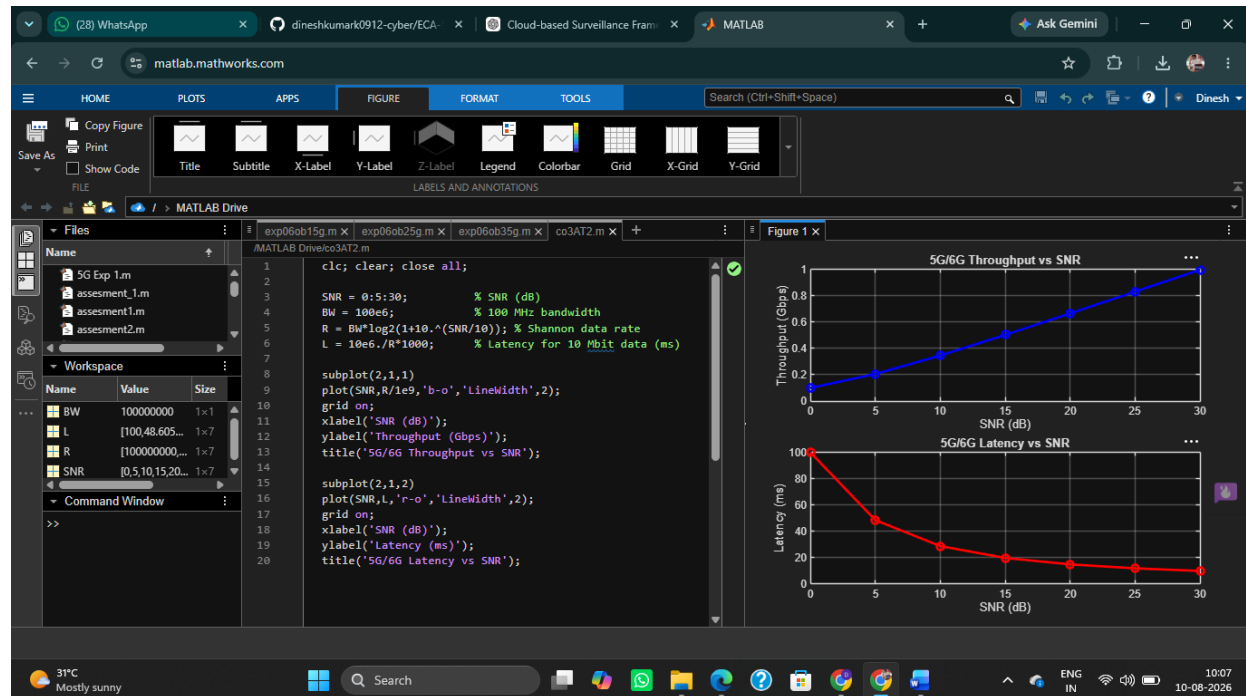
```
clc; clear; close all;
```

```
SNR = 0:5:30;      % SNR (dB)
BW = 100e6;        % 100 MHz bandwidth
R = BW*log2(1+10.^(SNR/10)); % Shannon data rate
L = 10e6./R*1000;  % Latency for 10 Mbit data (ms)
```

```
subplot(2,1,1)
plot(SNR,R/1e9,'b-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Gbps)');
title('5G/6G Throughput vs SNR');
```

```
subplot(2,1,2)
plot(SNR,L,'r-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Latency (ms)');
title('5G/6G Latency vs SNR');
```

Output:



Result:

Thus, a cloud-based wireless framework for smart surveillance was successfully proposed and simulated using MATLAB. The simulation shows that increasing SNR improves wireless throughput and reduces transmission latency. The proposed 5G/6G-based framework can therefore support high-data-rate, low-latency surveillance applications such as real-time video monitoring, cloud storage, and remote security monitoring.

5. Design an edge AI workflow for intelligent wireless data filtering.

Aim:

To design and simulate an Edge AI-based wireless data filtering system that removes low-value/noisy wireless data at the edge device before transmitting the useful information through the 5G/6G network.

Program Code:

```
clc;

clear;

close all;

t = 1:100;

data = sin(0.15*t) + 0.35*randn(1,100);

% Edge AI filtering decision

filtered = data;

filtered(abs(data) < 0.8) = 0;

plot(t,data,'k','LineWidth',1);

hold on;

plot(t,filtered,'b','LineWidth',2);

grid on;

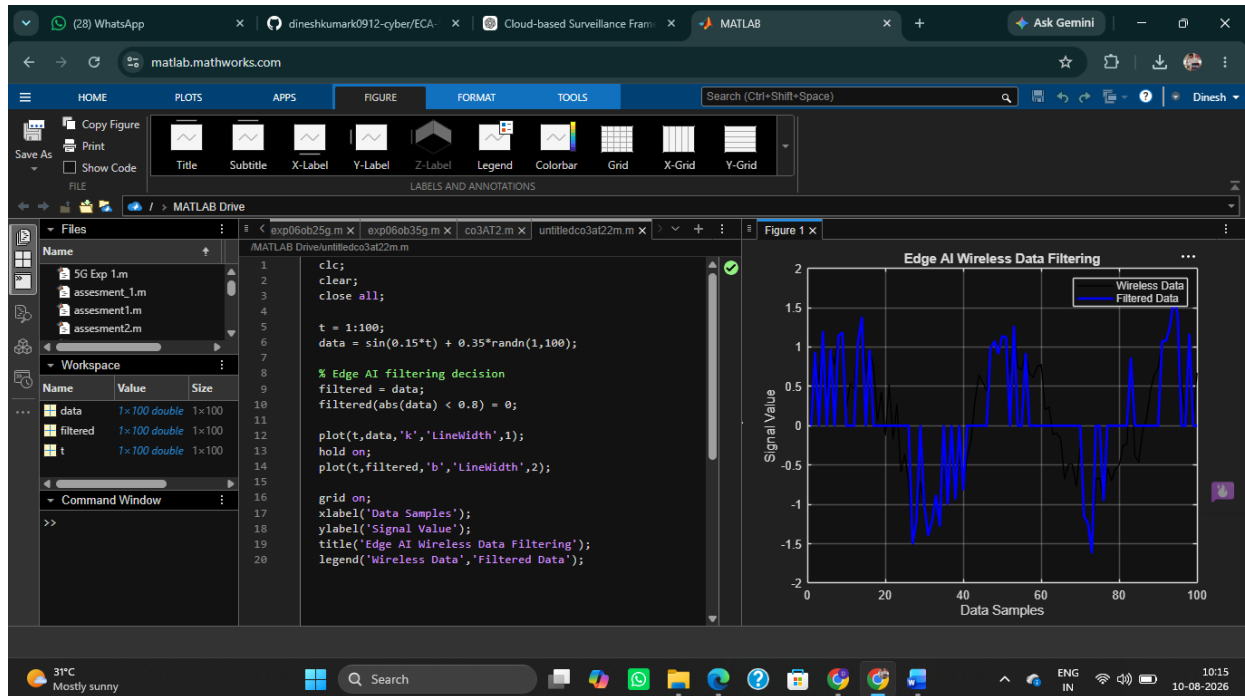
xlabel('Data Samples');

ylabel('Signal Value');

title('Edge AI Wireless Data Filtering');

legend('Wireless Data','Filtered Data');
```

Output:



Result:

The simulation demonstrates that Edge AI can filter low-value wireless data locally before transmission. This reduces unnecessary network traffic and can improve bandwidth utilization and communication efficiency in 5G/6G wireless networks.